Integral Inequalities

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The central idea. Many inequalities for finite sums continue to hold when sums are replaced with integrals. The main challenge is to recognize the right structure: a square, a product, a convex function, or a pair of similarly ordered functions. Equality cases are often as important as the inequality itself.
Unless otherwise specified, functions are real-valued and continuous on the indicated interval.

1. Nonnegativity and Completing the Square

Proposition 1.1: Nonnegativity of squares. *For any continuous real-valued function h on $[a, b]$,*

$$\int_a^b h(x)^2\,dx \geq 0,$$

with equality if and only if $h(x) = 0$ for every $x \in [a, b]$.

The principle becomes powerful when we can rewrite a complicated expression as a constant minus the square of another expression.

Example 1.2: A sharp quadratic bound. For any continuous real-valued function f on $[0, 1]$, show that

$$\int_0^1 f(x)(x - f(x))\,dx \leq \frac{1}{12},$$

and determine when equality holds.

Solution. Complete the square pointwise:

$$f(x)(x - f(x)) = \frac{x^2}{4} - \left(f(x) - \frac{x}{2}\right)^2.$$

Thus

$$\int_0^1 f(x)(x - f(x))\,dx = \frac{1}{12} - \int_0^1 \left(f(x) - \frac{x}{2}\right)^2 dx \leq \frac{1}{12}.$$

Equality holds precisely when $f(x) = x/2$ for all $x \in [0, 1]$. □

2. The Cauchy–Schwarz Inequality

Theorem 2.1: Integral Cauchy–Schwarz. *For square-integrable real-valued functions f and g on $[a,b]$,*

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \le \left(\int_a^b f(x)^2\,dx\right)\left(\int_a^b g(x)^2\,dx\right).$$

Equality holds when f and g are proportional almost everywhere, including the case where one is zero.

Proof. For any $t \in \mathbb{R}$,

$$0 \le \int_a^b (f - tg)^2 dx = \int_a^b f^2 dx - 2t\int_a^b fg\,dx + t^2\int_a^b g^2 dx.$$

If $\int_a^b g^2 dx > 0$, this quadratic in t must have nonpositive discriminant, yielding the inequality. If $g = 0$ almost everywhere, the result is immediate. □

Corollary 2.2: A fixed integral gives an L^2 bound. *Writing $L = b - a$ and taking $g \equiv 1$,*

$$\int_a^b f(x)^2\,dx \ge \frac{1}{L}\left(\int_a^b f(x)\,dx\right)^2.$$

Equality for continuous f occurs precisely when f is constant.

Corollary 2.3: Weighted Cauchy–Schwarz. *If $w(x) > 0$ on $[a,b]$, then*

$$\int_a^b \frac{f(x)^2}{w(x)}\,dx \ge \frac{\left(\int_a^b f(x)\,dx\right)^2}{\int_a^b w(x)\,dx}.$$

Equality occurs if and only if $f(x) = c\,w(x)$ for a constant c.

Example 2.4: Optimizing a weighted integral. Suppose $\int_0^1 f(x)\,dx = 1$. Find the minimum of

$$\int_0^1 \frac{f(x)^2}{1+x^2}\,dx.$$

Solution. Use $w(x) = 1 + x^2$. Then

$$\int_0^1 \frac{f(x)^2}{1+x^2}\,dx \geq \frac{1}{\int_0^1 (1+x^2)\,dx} = \frac{3}{4}.$$

Equality requires $f(x) = c(1+x^2)$. The integral constraint gives $c = 3/4$, so the minimum is $\boxed{3/4}$. □

3. Jensen's Inequality and Convexity

Definition 3.1: Convexity. A function ϕ on an interval is *convex* if

$$\phi(\lambda u + (1-\lambda)v) \leq \lambda\phi(u) + (1-\lambda)\phi(v)$$

for all u, v in its domain and $0 \leq \lambda \leq 1$. For twice differentiable ϕ, the condition $\phi'' \geq 0$ implies convexity. A *concave* function satisfies the reverse inequality.

Theorem 3.2: Jensen's integral inequality. *Let $L = b - a$ and suppose ϕ is convex on an interval containing the range of f. Then*

$$\phi\left(\frac{1}{L}\int_a^b f(x)\,dx\right) \leq \frac{1}{L}\int_a^b \phi(f(x))\,dx.$$

Reverse the inequality if ϕ is concave. If ϕ is strictly convex or strictly concave, equality for continuous f requires f to be constant.

Remark 3.3: Why Jensen works. Let $m = L^{-1}\int_a^b f(x)\,dx$. For a differentiable convex ϕ, its tangent line at m lies below its graph:

$$\phi(t) \geq \phi(m) + \phi'(m)(t-m).$$

Set $t = f(x)$ and integrate. The average of $f(x) - m$ is zero.

Example 3.4: A concave logarithm. If $f \geq 0$ and $\int_0^1 f(x)\,dx = 1$, prove

$$\int_0^1 \log(1+f(x))\,dx \leq \log 2.$$

Solution. The function $\phi(t) = \log(1+t)$ is strictly concave for $t \geq 0$ because $\phi''(t) = -(1+t)^{-2} < 0$. Jensen gives

$$\int_0^1 \log(1+f(x))\,dx \leq \log\left(1+\int_0^1 f(x)\,dx\right) = \log 2.$$

Equality holds exactly for $f \equiv 1$. □

Corollary 3.5: A useful power inequality. *For $p > 1$, convexity of $|t|^p$ implies*

$$\int_a^b |f(x)|^p\,dx \geq L^{1-p}\left|\int_a^b f(x)\,dx\right|^p.$$

This also follows from Hölder's inequality.

4. Chebyshev's Integral Inequality

Theorem 4.1: Chebyshev's integral inequality. *If f, g are both nondecreasing or both nonincreasing on $[a,b]$, then*

$$L\int_a^b f(x)g(x)\,dx \geq \left(\int_a^b f(x)\,dx\right)\left(\int_a^b g(x)\,dx\right), \qquad L = b-a.$$

If one function is nondecreasing and the other is nonincreasing, the inequality reverses.

Proof. When f and g are similarly ordered, $[f(x)-f(y)][g(x)-g(y)] \geq 0$. Integrate over $[a,b]^2$ and expand:

$$0 \leq \int_a^b\int_a^b [f(x)-f(y)][g(x)-g(y)]\,dx\,dy = 2L\int_a^b fg - 2\left(\int_a^b f\right)\left(\int_a^b g\right).$$

Rearranging proves the claim. The opposite-monotonicity case follows by reversing the pointwise inequality. □

Example 4.2: Monotonicity creates a lower bound. If f is nondecreasing on $[0,1]$, prove

$$\int_0^1 xf(x)\,dx \geq \frac{1}{2}\int_0^1 f(x)\,dx.$$

Solution. Apply Chebyshev to $g(x) = x$. Then

$$\int_0^1 xf(x)\,dx \geq \left(\int_0^1 x\,dx\right)\left(\int_0^1 f(x)\,dx\right) = \frac{1}{2}\int_0^1 f(x)\,dx.$$

Because x is strictly increasing, equality for continuous nondecreasing f requires f to be constant. □

5. How to Choose a Technique

- **An average and a square:** try Cauchy–Schwarz with $g \equiv 1$.
- **An expression f^2/w:** try weighted Cauchy–Schwarz.
- **A convex or concave function of f:** try Jensen.
- **Two monotone factors:** try Chebyshev.
- **A quadratic expression in f:** try completing the square.

Always check the equality case. It tells you both whether the estimate is sharp and what the extremizing function should be.

6. Practice Problems

The problems are ordered approximately by difficulty. Unless specified otherwise, functions are continuous on $[0, 1]$. *Determine equality cases whenever an inequality is given.*

6.1. Foundations

Problem 1: Fixed average. Suppose $\int_0^1 f(x)\,dx = 2$. Prove

$$\int_0^1 f(x)^2\,dx \geq 4.$$

Problem 2: A weighted minimum. Suppose $\int_0^1 f(x)\,dx = 1$. Prove

$$\int_0^1 \frac{f(x)^2}{1+x}\,dx \geq \frac{2}{3}.$$

Problem 3: Power integrals. Let $a, b, t > 0$. Compare

$$P = \left(\int_0^t x^{a+b}\,dx\right)^2, \qquad Q = \left(\int_0^t x^{2a}\,dx\right)\left(\int_0^t x^{2b}\,dx\right).$$

Which is greater, and when are they equal? *(Adapted from Cambridge NRICH.)*

6.2. Choosing the Inequality

Problem 4: Jensen in disguise. Assume $f(x) \geq 0$ and $\int_0^1 f(x)\,dx = 1$. Show

$$\int_0^1 \frac{1}{1+f(x)}\,dx \geq \frac{1}{2}.$$

Problem 5: A reciprocal product. Suppose $f(x) > 0$ on $[0,1]$. Prove

$$\left(\int_0^1 xf(x)\,dx\right)\left(\int_0^1 \frac{x}{f(x)}\,dx\right) \geq \frac{1}{4}.$$

Problem 6: A monotonicity constraint. If f is nondecreasing, prove

$$\int_0^1 x^2 f(x)\,dx \geq \frac{1}{3}\int_0^1 f(x)\,dx.$$

Problem 7: Reflection. For a real-valued continuous function f, show

$$\int_0^1 f(x)f(1-x)\,dx \leq \int_0^1 f(x)^2\,dx.$$

6.3. Challenges

Problem 8: Two moment constraints. Suppose f is continuous and

$$\int_0^1 f(x)\,dx = 1, \qquad \int_0^1 xf(x)\,dx = 1.$$

Prove $\int_0^1 f(x)^2\,dx \geq 4$. *(From* Putnam and Beyond*, Problem 475.)*

Problem 9: From derivatives to functions. Suppose u is continuously differentiable and $u(0) = 0$. Prove

$$\int_0^1 u(x)^2\,dx \le \frac{1}{2}\int_0^1 u'(x)^2\,dx.$$

Hint: Write $u(x)$ as an integral of its derivative.

Problem 10: An olympiad functional equation. Determine all continuous $f : [0,1] \to \mathbb{R}$ satisfying

$$\int_0^1 f(x)\,dx = \frac{1}{3} + \int_0^1 f(x^2)^2\,dx.$$

(1982 Romanian Mathematical Olympiad, as presented in Putnam and Beyond*.)*

Sources for Selected Problems

Cambridge NRICH, *Integral Inequality* (Problem 3), NRICH online problem. R. Gelca and T. Andreescu, *Putnam and Beyond* (Springer, 2007), integral inequalities section (Problems 8 and 10). Other problems are standard or independently formulated.